Valparaíso, October 28, 2026

Editor
European Journal of Operational Research

Dear Editor,

Please find enclosed my manuscript "Fast exact overlap minimization for three-dimensional irregular strip packing of polyhedra", which I submit for consideration for publication in the European Journal of Operational Research.

The paper addresses the three-dimensional irregular strip packing problem, in which polyhedra, possibly non-convex, are placed by translation in a container of fixed base so as to minimize its height. In two dimensions, the best methods for irregular strip packing are overlap-minimization heuristics, but in three dimensions the existing approaches either use an expensive exact overlap measure or approximate the pieces with voxels. The main contribution is a fast exact move for polyhedra represented as unions of convex parts: the overlap of two parts is measured by the penetration depth in their no-fit polytope, and the best position of a piece along an axis is found exactly by evaluating a few breakpoints. Around this move, the paper builds a simple method that adapts known components from two-dimensional nesting, and analyzes which of them matter.

On the classical benchmarks of polyhedral pieces, the method improves the best known heights of 11 of 13 instances within two minutes, reaches in 10 seconds the quality that the previous exact local search attains in 250 seconds (normalized to the same hardware), and matches 24 of the 27 proven optima of the small instances of recent mixed-integer models. It also reports the limitations of the method on curved meshes, where voxel-based approaches remain superior. The source code, the solutions and the scripts that reproduce all the experiments are publicly available at `https://github.com/rilianx/poly3d-packing`.

I believe the paper fits the scope of the journal, which has published most of the recent work on irregular packing, including the references compared with. The manuscript has not been published and is not under consideration elsewhere. There are no conflicts of interest to declare.

Sincerely,

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